From dispersed knowledge to ontology: A proposal for formalising and integrating 4D printing in design

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Abstract. Current research efforts related to 4D printing (4DP) has reached an unprecedented level in smart materials (SM) driven era. It is mainly about manufacturing parts/products combining additive manufacturing (AM) techniques and reactive matters. The latter association leads to transformable products presenting abilities of shape morphing, colour changing, etc., after being triggered with adequate stimuli. Thus, many ad-hoc solutions appeared in literature, in the last decade, with different dispersed expertises. Indeed, the information available about how to carry such smart products automatically is limited and poorly known. This lack of common core knowledge is primarily due to the various skills involved in printing stimuli-reactive products. Hence, different user profiles, i.e. the product designer, product architect, process planner, etc., cannot profit from this pioneering technology. Thus, the product design process must incorporate such technology to tackle design for 4DP issues. In this context, this paper proposes a framework that starts from establishing an ontological structure for the semantic and logical description of smart products respecting multi-dimensionalism (3D &4D), semantic reasoning theories and description logic language. It deploys, in the end, a strategy for integrating the mentioned formalised knowledge in cognitive CAD methods and tools.

Keywords: Ontology, Semantics, Smart Products, Knowledge, Product Design Process, Design for 4D Printing.

1. Introduction

4D printing (4DP) is gaining increasing interest due to the smart materials (SM) capabilities and additive manufacturing (AM) techniques. This technology associates both competencies to deliver smart part/product that is sensitive to specific stimuli, so, transformation capabilities are incorporated into the manufactured products. Despite still being in its infancy phase, 4DP generates considerable interest due to its crucial role in manufacturing transformable and adaptive structures conducting an estimable change [1]. Combining AM techniques with SM provides a temporal dimension (t) to the three geometrical dimensions (x,y,z). Last decade was considered to be the triggering period of this innovative technology. Preliminary studies about 4D printable structures involve multi-material printing possibilities [2] to correlate inert to smart materials. Hence, the products are designed and manufactured as living components capable of moving to various physical states or geometric arrangements concerning their trigger conditions [3]. Triggers, i.e. stimuli, are needed to activate shifts at multiple levels, e.g. structure, form, property and functionality. For all the stakeholders mentioned above, 4DP is considered a promising industrial technology for the next generation of live products [4]. So that, the next decade is about to witness a significant ascent in designing and manufacturing transformable and reactive products. But most of the previous works have only focused on multiplying the used materials, stimuli, AM processes and the possible transformations with neglecting the need to theorise the knowledge behind. For that reason, the dispersed knowledge must be formalised in rigorous understanding; therefore, product designers, architects and process planners can significantly restate it in part/product design phases. Many researchers were interested in the concept of design for excellence DFX [5], which attempts to concurrently scrutinise product lifecycle using assembly, processing, recycling, and mainly other product features.

Similarly, design for 4DP problems needs to be tackled, i.e., developing knowledge models, methods, and tools to optimise a design for 4D printable product. Actual 4DP experiments and related proofs-of-concept are introducing disparate knowledge with interdisciplinary expertise. Consequently, the research goal is set to emphasise and federate 4D printing-related conceptual knowledge involving AM processes and techniques, SMs, stimuli and transformation problems. Hence, an ontology is implemented for this purpose thus the knowledge-behind is computer-readable and reused at different design stages of the product. In other words, product architects, designers and manufacturers are supported to make the best choices, in the future, during the design phase of living and transformable objects. In this paper, an ontology is proposed to ensure the conceptual representation and logical description of transformable objects in the age of 4D printing. Besides, this ontology is formalised and presented in description logic (DL) language in order to demonstrate the possible knowledge extraction for each

user (i.e. designer, process planner, etc).
This paper is organised as follows: the second section set up some relevant research literature. The third section proposes a framework for developing the proposed ontology and presenting the possible usage scenarios after implementing a domain ontology that serves as a "database" for 4DP knowledge. Moreover, it presents the formalisation of the ontology using DL language. The fourth section illustrates a case study for demonstration; finally, conclusions are stated, and future work is addressed in the last section.

2 Background

2.1 4D printing

Transformability and product changeability during or post manufacturing were already addressed in the last two decades, before the term 4DP has been coined. Previously, this research area was discussed by polymer scientists with consideration to form morphing structures and shape memory polymers (e.g. artificial muscles) [6-7]. Preliminary studies of polymers have considered behavioural studies, glass transition temperature, stress magnitude and elasto-viscoplasticity deformation [8-10], with widespread usages through various domains [11-15]. Along with the cited works, particular substance embedding dynamic behaviours have been developed to accomplish primitive transformations such as bending, twisting and folding – until exposed to solvent, light, magnetic and electrical fields [16-17]. Besides, shape-memory composites and pH-responsive materials like hydrogels were developed for self-folding transformation, presented in many experiments using origami hinges or multi-layered materials[18-20]. Accordingly, the above research progress has initiated programming materials by tuning constitutive laws [21]. As a result, 4DP was coined to cover these attempts of AM technologies, particularly material jetting [22-23]. Much of their proposed study deal with ad-hoc works involving multi-materials, immediately from the printing bed, dipped in water (e.g. hydrophilic silicone material) or triggered by electricity. A growing body of literature has examined folding applications so that researchers have shown several shifting structures based on different stimuli, structures and material distributions [24]. Thus, 4DP allows transformation based sequence over spatial dimensions (i.e. 2D and 3D) from one state to another in terms of structure, properties and functions [25-26]. Furthermore, [27] illustrated a linear stretching structure (from 1D to 1D), a 2D folding and stretching structure, and a 3D topographical surface transition. Then multiple transformation-dimensional possibilities were demonstrated, and numerous study niches were opened. Other researchers have based their efforts on manipulating bending and folding primitives by using reverse engineering and measuring such parameters such as fibre weight, hinge length and position, strain, temperature, deformation and activation processes [28-30]. [31] reviewed bending structures' kinematic analyses and their corresponding folding patterns triggered by multiple factors (e.g. thermal, chemical, etc.). Similarly, diverse plastic materials have been developed to fuel advancement in 4DP, such as gels with photo-responsive fabrics, thin composites and polystyrene [32]. Henceforward, it can be estimated that the fundamental objects constructing 4DP technology are triggers, geometry, AM processes and techniques, SMs and transformation functions [2]. Merely all the experiments have used proof-of-concept machines/devices, and many AM methods, ranging from material extrusion (fused filament production and direct ink writing), photopolymerisation (stereolithography, digital lighting processing) to material jetting (drop-on-demand) [33-34]. Here on, advances are becoming more mature, like the research works of [35-36] in which, they determine SM distributions at the early design stage of 4D printed structures by considering a voxel-based simulation approach coupled with genetic algorithms.

The above works provide plenty of interrelated information and unstructured knowledge. Then, it shows the complexity of integrating 4DP in a complete product life cycle, i.e. design, simulation, production, etc. Besides, designers and architects still can't take profits from this innovative paradigm. So, capturing the knowledge behind is crucial at this stage.

2.2 Data formalisation-oriented tool: Ontology and DL language

Ontology is defined as a "database" – which current topics, objects, processes and theories as entities or concepts and relationships as unary or binary predicates – designed with axioms, definitions and theorems [37-39]. As a tool for capturing knowledge and formalising it, ontology was introduced recently in mechanical engineering. Hence, a feature-based ontology was developed to gather semantic knowledge in order to integrate computer-aided design (CAD) and computer-aided process planning [40]. The interoperable protocol was to communicate existing and following generation CAD programs. [41] have proposed an ontology to support the preparation of the first-order process for sheet lamination technology. For the same manufacturing process, [42] developed the ontology

embedding manufacturing rules for automated feature identification technique. An ontology dedicated to designing AM was suggested in order to produce a system for component design details, AM processes and the ability to represent design rules [43]. It is then possible to test design features against production restrictions. Similarly, an informal model of generative ontology was developed for AM method planning [44]. Zhang et al., (2015) provided An ontological system for selecting and semantically extracting materials in domain-oriented information from a material viewpoint was provided by [45]. Hagedorn *et al.* (2019) [46] showed the use of basic formal ontology (BFO) over-engineering realms in the interoperation, inquiry, and parallel acquisition of information. Moreover, an ontology has been built to formalize 4DP knowledge at a product/part level by basic formal ontology BFO [47] On the other hand, description logics are formal representation languages that have been itiniated with Karlsruhe and are decidable fragments of first-order logic [48]. It is of particular importance in providing logical formalism for quality ontologies. In DL, a distinction is drawn between the so called T-Box (terminological level) and the A-Box (assertional level). A DL terminology is a set of terminological axioms, consisting of terminological constructions of ontological elements. Ontological elements are of two types: non-relational and relational elements. Non-relational elements are concepts (classes) and datatypes, and relational elements are roles and attributes.Hence, ontologies and DL language are essential for implementing a preliminary design phase for 4DP and describing the spatiotemporal phenomena behind these transformable structures.

2.3 Design for 4DP issues

The wide spectrum of research initiatives in mechanical design world [49-51] and the current limitations of 4D printing demonstarate that tackeling a proactive design process for this technology is a research area in a full expansion. Indeed, interdisciplinary issues in product design and development have been addressed in the past especially in the product development phase where design activities and decisions have a great impact on downstream processes (i.e. manufacturing, assembly, maintenance, etc) [52-53]. Nonetheless, the recent implementations of design for X - in which X could be for example assembly or manufacturing - have led to enormous benefits including simplification of products and reduction of costs [54-56]. The latter is encouraging to propose a design for 4D printing roadmap that can starts eventually with a design for 4DP framework that promotes a multipe involved stakeholders (i.e. product architect, product design, material expert, etc.) and various 4DP components (i.e. materials, stimuli, processes, transformations, etc.).

As a conclusion and with regard to aforementioned research works, it seems crucial to propose an ontology federating the 4DP knowledge and to formalise it, using DL language in order to apply a top-down design strategy and to propose knowledge based design for 4DP approach.

3 Contribution

3.1 Framework

The framework presented below in Fig.1 tries to establish a comprehensive method of formalising and describing the knowledge qualitatively behind 4DP (symbolic learning). The 4DP ontology is implemented with refer to multiple proof-of-concepts (ad-hoc research experiments) and data sources, i.e., AM processes and techniques standards, materials databases. Nevertheless, it is crucial to address and acquire fundamental theories like *multi-dimensionalism* since the research gap is handling transformable objects over time, *semantic reasoning* to cover the articulation between immaterial and material objects, *mereotopology* to address spatial relationships (parts and wholes relationships) and *changeability management* so that the ontology in continually updated.
Once the classes, data- properties and instances are set, the ontology is modelled within distributed profiles to facilitate its use, i.e., the product designer, the product architect, the process planner, the material expert. Based on DL language, the ontology is presented formally, and the above cited profiles are defined to facilitate knowledge extractions and ontology understanding.

This approach to design for 4DP is moving from a strict sequential organisation logic of marketing/study/industrialisation/manufacturing actors to an integrated sense of simultaneous cooperation between all the product actors. The objective of these new organisational methods is to design concurrently a product efficient and harmonious on the different axes delineated by the main actors' needs.

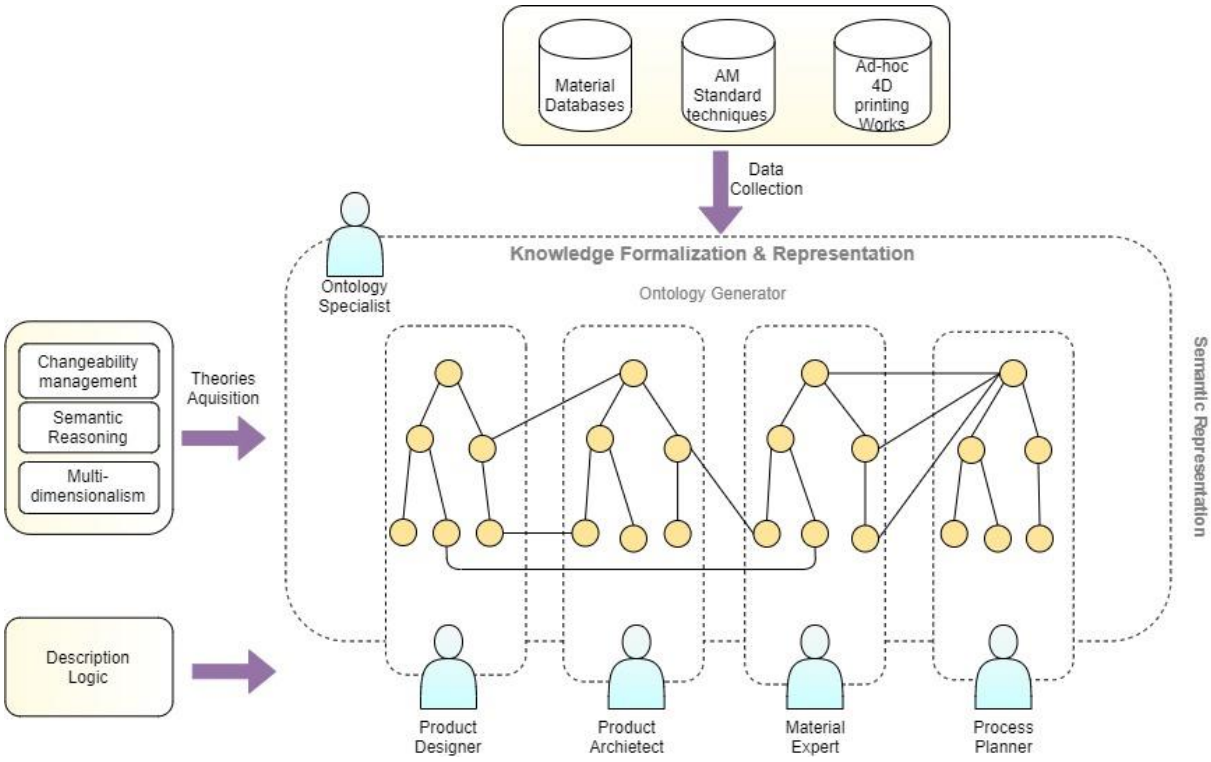

Fig. 1. Framework

3.2 Ontology Implementation

Our ontology is built upon top-down and bottom-up constructions strategies. The main goal is to formalise and describe a design for 4D printing knowledge, which needs mapping concepts and properties as presented in Fig.2 and Fig.3 using Protégé 5.5 software. As mentioned in the litterature, the ontology was implemented respecting A-box and T-box: grey rectangles stand for the classes and hatched lines present mainly the object properties between classes. The inheritance is shown in Fig.4 by citing the different subclasses of some concepts. A product can represent here the 4D printed part within its multiple states. The Embedded_part is a separate manufactured part generally added while the AM process to trigger the transformation. Homogeneous_part is printed with exclusively one material, however, the Hybrid_part is printed with at least two materials. The Voxel class is an elementary volume arranged and spaced in a Voxel_grid which has different data properties. It enables to decompose a 3D object in multiple sub-objects.

Then, the Transformation_function(s) are capitalised into six as demonstrated in the Fig.4(c) below. Chemical_properties_changing is about a shifting in colour, mechanical properties ...etc. And State_changing stands for the changement of the material state (liquid/solid). Transformation_dimension captures the dimensional natures of the transformations' states (e.g., 1D to 1D, 1D to 2D, 1D to 3D, 2D to 2D, 2D to 3D, 3D to 3D). Transformation_effect presents the cycle of transformability like One_way which the change of the original shape/ property to a permanent one. Also, we can invoke Reversible (i.e. two ways) and Repeatable (i.e. more than two ways). Transformation_function is fulfilled by a Transformation_process which has a set of Transformation_step(s) (i.e. Training which limited to one Consyrained_thermo_mechanics mechanism, Stimulation which consists in applying the trigger(s) and Actuation is the main act of shifting that a 4D printed part fulfils (see Fig.4(d)).

Besides, a material class can have Mechanical_property(ies), Technical_function (see Fig.4(f)) and distributed with specific Material_distribution (see Fig.4(e)). The latter exhibits spatial distribution patterns of smart (i.e. Active) and inert (i.e. Structural or Sacrificial) Materials. Stimulus class englobes the numerical and qualitative information that a Stimulus can bear to trigger such a material and consequently a 4D printed part as listed in Fig.4(b). Therefore, a stimulus triggers the Active_material and the _Object_aggregate, affects the Geometry and the Mechanical_property and enables the Transformation_function.

Last but not least, Additive_manufacturing_process class is already formalised by the "American society for testing and materials" and listed as shown in Fig.4(a). And in the same context, Process_condition class specifies machine control parameters.

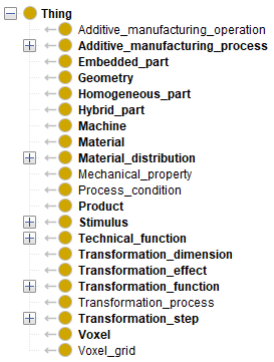

Fig. 2. Upper Classes defined in Protégé software

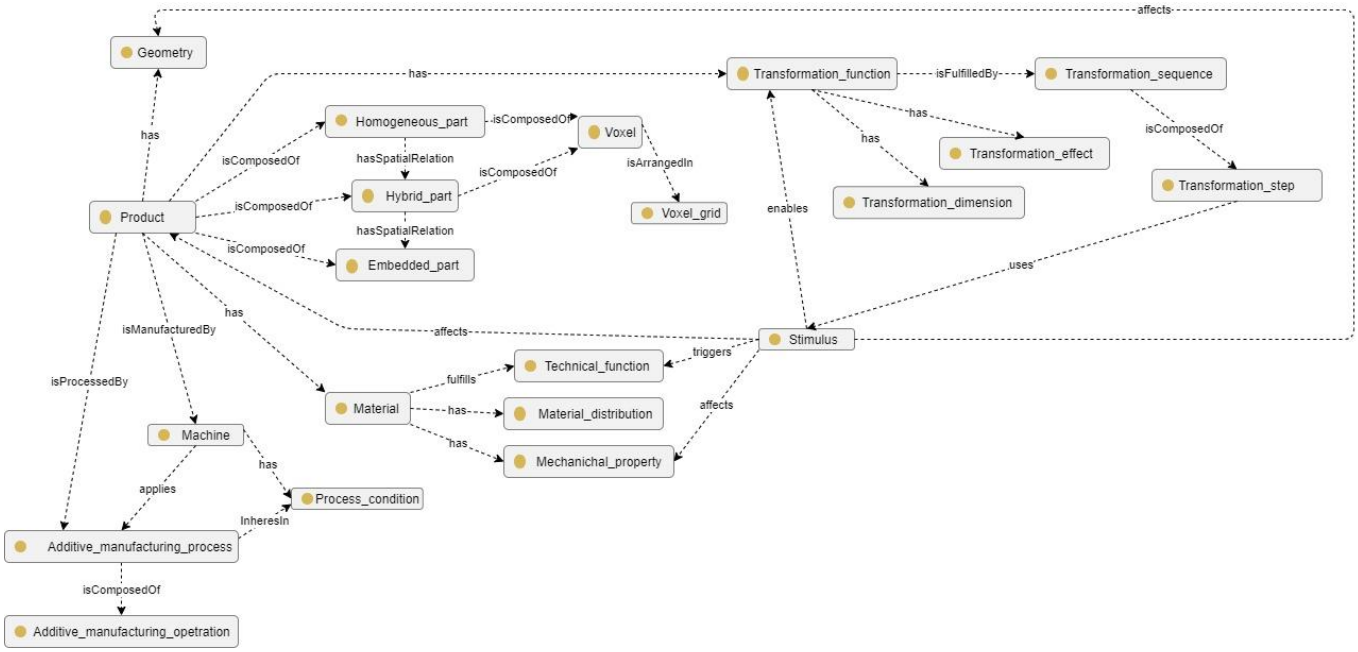

Fig.3. The proposed ontology-- for capitalising 4DP knowledge - graph

Fig.4. Subclasses of various classes defined in the ontology

3.3 Desciption Logic Language

For the sake of clarity, the ontology is formalised within the need of the four user profiles presented in the framework in the Fig.1. following the definitions below:

- ✓ Top classes are listed in *Concept Set* s_{ci}
- ✓ Object properties are listed in *Concept Subsumption* $\leq_{ci}$
- ✓ Data types are listed in *DataType Set* s_T
- ✓ Data attributes are listed in *DataAttribute Set* s_A
- ✓ Data properties are listed in *DataProperties Set* s_A

As for the product designer:

Concept Set s_{c1} *= {TopConcept, Product, Homogeneous_part, Hybrid_part, Voxel, Voxel_grid, Geometry, Transformation_function, Embedded_part}*

Concept Subsumption $\leq_{c1}$ *={(Top_concept, Product), (Top_concept ,Homogeneous_part), (Top_concept ,Hybrid_part), (Top_concept, Voxel), (Top_concept ,Voxel_grid) , (Top_concept , Geometry), (Top_concept, Transformation_function), (Top_concept, Embedded_part), (Embedded_part, Film), (Embedded_part, Electronic_component), (Object, Voxel), (Object, Embedded_part), (Object_aggregate, Hybrid_part), (Object_aggregate, Homogeneous_part), (Object_aggregate, assembly),* $(\cap_C$*(Voxel,* $\rho_{\forall R}(isArrangedIn,$ *Voxel_grid), Object)),* $(\cap_C\ (\rho_{\exists R}(has,\ Object), Geometry))$,
$(\cap_C\ (\rho_{\exists R}(has,\ Object),\ Transformation_function))$,
$(\cap_C\ (\rho_R(isComposedOf,\ Homogeneous_part),\ Voxel))$,
$(\cap_C\ (\rho_R(isComposedOf,\ Hybrid_part),\ Voxel))$,
$(\cap_C\ (\rho_R(isComposedOf,\ Assembly), Homogeneous_part))$,
$(\cap_C\ (\rho_R(isComposedOf,\ Assembly), Hybrid_part))$,
$(\cap_C\ (\rho_R(isComposedOf,\ Assembly), Embedded_part))$,
$(\cap_C\ (\rho_R(hasSpatialRelation, Homogeneous_part), Hybrid_part))$,
$(\cap_C\ (\rho_R(hasSpatialRelation, Hybrid_part), Embedded_part))$,

$(\rho_{\forall A}(Position,$ *xsd: decimal), Voxel),* $(\rho_{\forall A}(Color,$ *xsd: string), Voxel),*

$(\rho_T(Initial_state,$ *xsd: string), Object),* $(\rho_T(Final_state,$ *xsd: string),Object)}*

DataType Set s_T *={TopDataType, Initial_state, Final_state}*

DataAttribute Set s_A *={TopDataAttribute, Color, Position}*

DataProperties Set s_A *={TopDataProperty, isComposedOf, hasSpatialRelation, has, isArrangedIn}*

As for the product Architect:

Set of top concepts s_{c4} *= { TopConcept, Transformation_function, Transformation_process, Transformation_step, Transformation_effect, Transformation_dimension}*

Concept Subsumption $\leq_{c4}$ *={(Top_concept, Transformation_function), (Top_concept, Transformation_process), (Top_concept , Transformation_step), (Top_concept, Transformation_effect), (Top_concept, Transformation_dimension), (Transformation_function, Bending),…, (Bending, Buckling), (Transformation_step, Training), … ,(Training, Constrained_thermo_mechanical),*
$(\cap_C$*(Transformation_function,* $\rho_R(isFulfilledBy,$ *Transformation_process))),*

$(\cap_C\ (Transformation_function,\ \rho_R(hasTransformationEffect,$ *Transformation_effect))),*

$(\cap_C$*(Transformation_function,* $\rho_R(hasTransformationDimension,$ *Transformation_dimension))),*

$(\cap_C$*(Transformation_function,* $\rho_R(isComposedOf,$ *Technical_function))),* $(\cap_C$*(Transformation_process* $\rho_R(isPartOf,$ *Transformation_step))),* $(\cap_C$*(Stimulation* $\rho_R(uses,$ *Stimulus))),* $(\rho_T(Bending_torque,$ *xsd: decimal), Bending),* $(\rho_T(Bending_radius,$ *xsd: decimal), Bending),* $(\rho_T(Twisiting_angle,$ *xsd: decimal), Twisting),* $(\rho_T(Folding_angle,$ *xsd: decimal), Folding),* $(\rho_T(Folding_torque,$ *xsd: decimal), Folding)}*

DataProperties Set s_A *={TopDataProperty, isPartOf, uses, isFulfilledBy, isComposedOf, hasTransformationDimension, hasTransformationEffect}*

DataType Set s_T*={Bending_Torque, Bending_radius, Twisting_angle, Folding_angle, Folding_torque}*

Set of Instances s_I *= { « 1D_to_1D »,*
« 1D_to_2D », « 1D_to_3D », « 2D_to_2D », « 2D_to_3D », « 3D_to_3D », « One_way », « Reversible »,

« Repeatable » }

Set of Concept Instantiation i_c *= { (Transformation_dimension, 1D_to_1D), (Transformation_dimension, 1D_to_2D), (Transformation_dimension, 1D_to_3D), (Transformation_dimension, 2D_to_2D), (Transformation_dimension, 2D_to_3D), (Transformation_dimension, 3D_to_3D), (Transformation_effect, One_way), (Transformation_effect, Reversible), (Transformation_effect, Repeatable) }*

4 Use Case: Deployable Satellite

A deployable satellite prototype with four solar panels executing an unfolding transformation from an initial folded state (Fig.5 (a)) to a final completely unfolded state (Fig.5(b)) is presented as a case study to demonstrate the applicability of the developed ontology. The consistency of ontology is checked in the protégé software using HermiT reasoner. It is about describing and formalizing the knowledge behind this 3D-to3D transformation.

The objective here to capture relevant knowledge ensuring the transformation from a Folded_structure to an Unfolded_structure. The ontology presented in the latter section is manually populated which has led to an instance graph represented in Fig.6. For sake of clarity and in order to avoid repetition, the deployable satellite ontology is presented in modular format. Since, it has four similar legs of solar panels which involves parts with active and inert materials, only one leg is described in Fig7. As a fragment of the module of Homogeneous and hybrid parts. The parts are topologically connected with isTangentTo relation. Also, the transformation steps of each smart hinge in one leg and of the four legs are quite similar, that's why the transformation process of only one leg of solar panels is captured in the Fig.8 as a fragment of the module of transformation steps. Each hinge will be transformed in an unfolded one. This prototype is printed through a FDM machine respecting a set of universal and agreed conditions and operations. The material used is sensitive to temperature variations thus the technical function of this unfolding is thermoactive. As such, voxel class is not considered here because the material distribution of both active (Panel0i_j) and smart (Hinge0i_j) parts is uniform, hence there no need to decrease the geometrical representative level.

Respecting the DL sets defined in the third sections, two views are manually extracted respecting the needs of the product designer (Fig.9) and product architect (Fig.10).

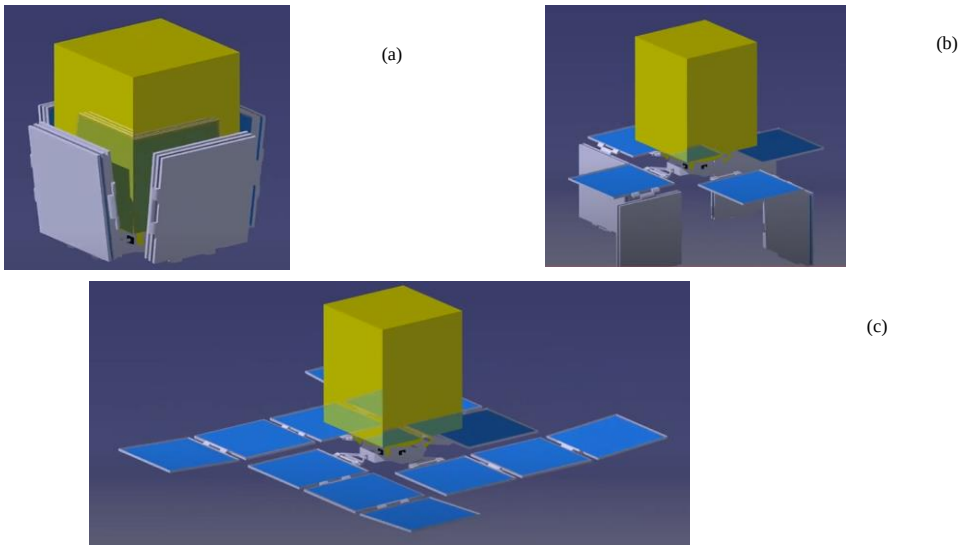

Fig.5. Deploying Satellite - Use Case (a) Initial State (b) Intermediary State (c) Folded State

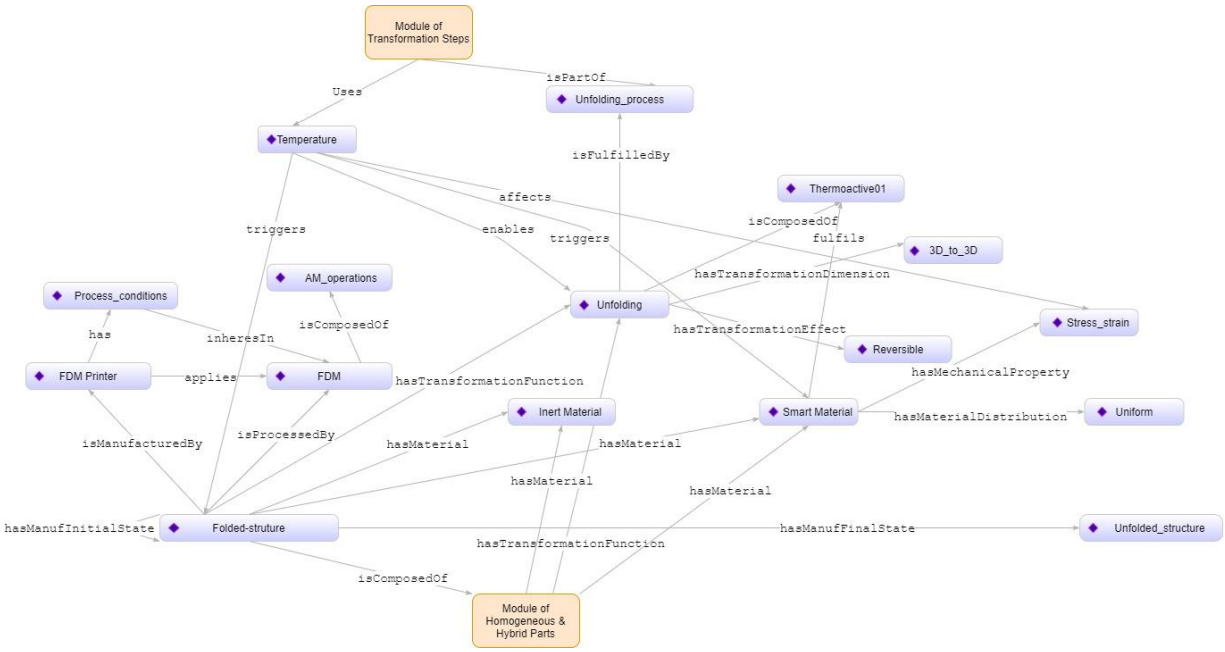

Fig.6. Intances graph capitalising the knowledge of 4DP in Deployable Satellite

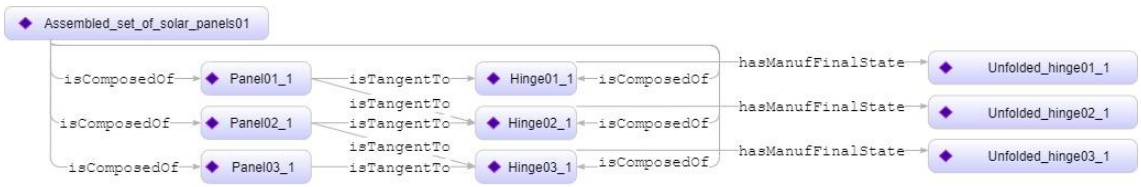

Fig.7. Fragment of the module of homogeneous & hybrid parts

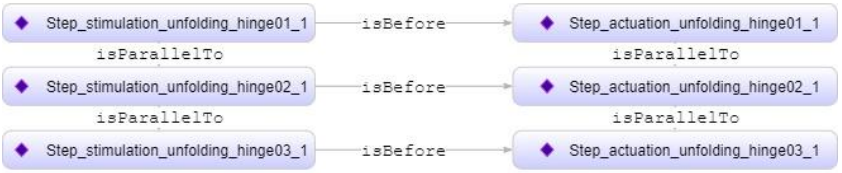

Fig.8. Fragment of the module of Transformation Steps

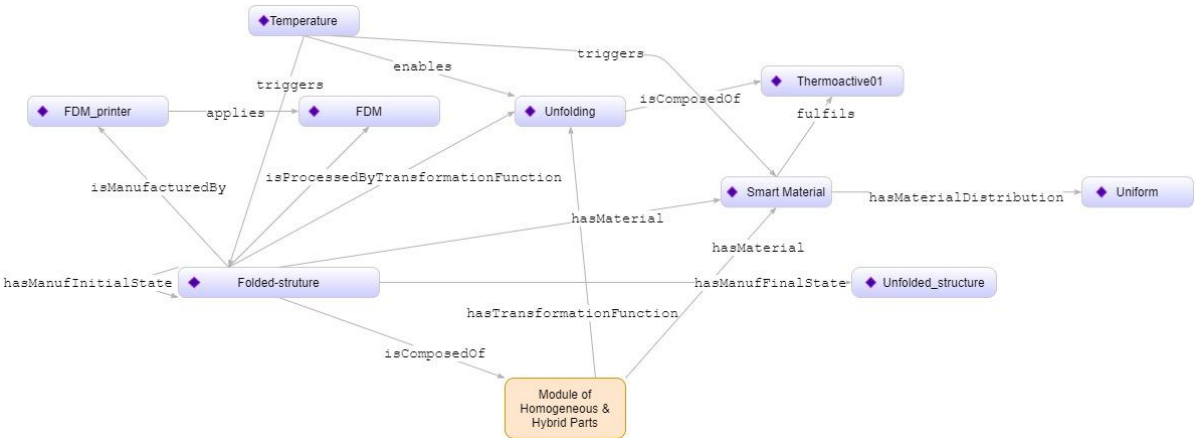

Fig.9. The design view for product designer

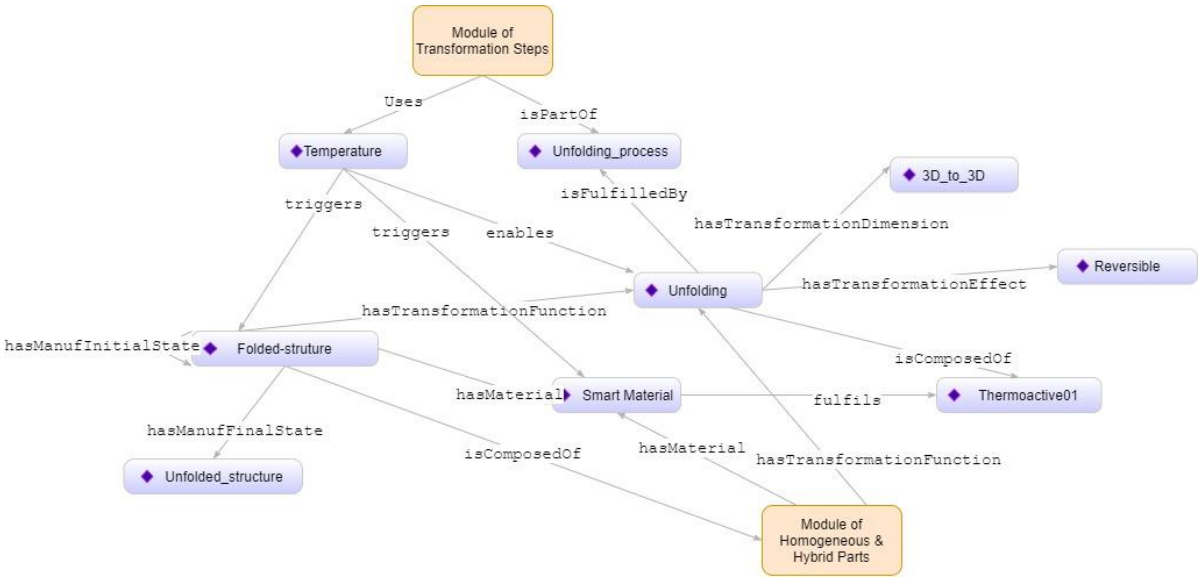

Fig.10. The transformation view for product architect

5 CONCLUSION & FUTURE WORK

Our research has stressed the formalisation and representation of 4DP knowledge relating AM processes and techniques, AMs, stimuli, and transformation matters. It therefore represents a pioneer effort towards knowledge federation and representation for multiple purposes aligned with a preliminary design phase. To do so, an ontological framework has been beforehand established with the underlying theories, along with the use of 4DP knowledge in design. The proposed ontology includes reasoning capabilities supported by DL rules. However, additional reasoning layers will be needed to fully capture 4DP knowledge over views.

For instance, future work is presented in Fig.11 as perspectives projecting possible approaches after implementing the 4DP ontology. A dedicated API is about to make the ontology and its whole axioms readable by a specific CAD software (e.g., Grasshopper as a plug-in of Rhinoceros). Hence, a structured library is ready for use for the different four actors with multiple purposes. A cognitive prediction of the transformations sequence is envisaged through ML tools based on the reuse of ad-hoc experiments as a learning data. Hence an appropriate CAD tool is ready to automatically generate 3D CAD geometric models, i.e., functional and fabricable three-dimensional geometric shapes. And then, the simulation engineer can take profit from the created geometric inference engine.

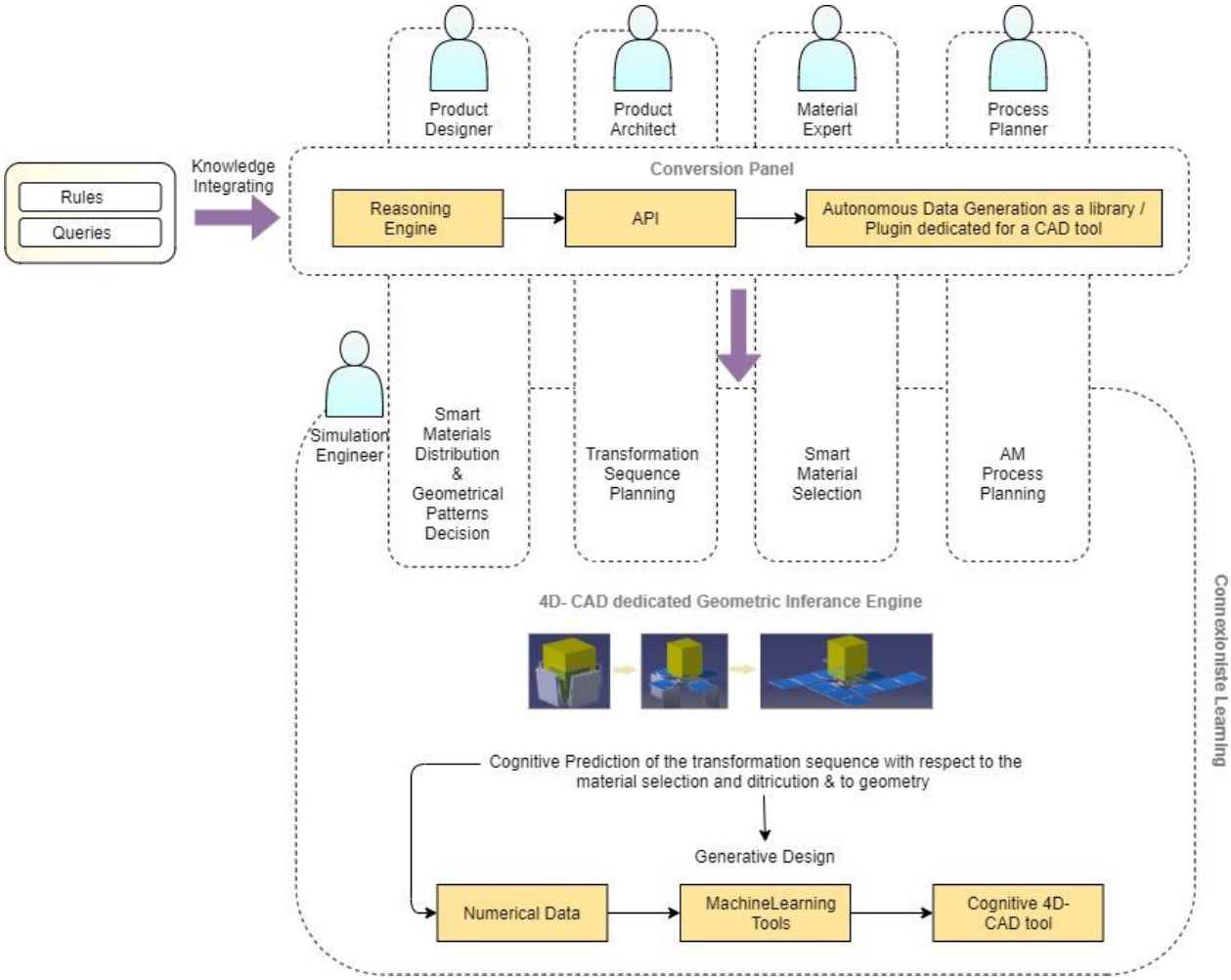

Fig.11. Proposed Perspectives

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References

1. Tibbits, S., McKnelly, C., Olguin, C., Dikovsky, D., & Hirsch, S. (2014). 4D printing and universal transformation. Proceedings of the 34th Annual Conference of the Association for Computer Aided Design in Architecture, 539–548.
2. Momeni, F., N, S. M. M. H., Liu, X., & Ni, J. (2017). A review of 4D printing. Materials & Design, 122, 42–79.
3. Choi, J., Kwon, O., Jo, W., Lee, H. J., & Moon, M.-W. (2015). 4D Printing Technology : A Review. *3D Printing and Additive Manufacturing*, *2*(4), 159–167.
4. Kuang, X., Wu, J., Chen, K., Zhao, Z., Ding, Z., Hu, F., … Qi, H. J. (2019). Grayscale digital light processing 3D printing for highly functionally graded materials. *Science Advances*, *5*(5), 1–10.
5. Holt, R., & Barnes, C. (2010). Towards an integrated approach to "' Design for X "': an agenda for decision-based DFX research. *Res Eng Design*, *21*, 123–136.
6. Kang, Y., Walish, J. J., Gorishnyy, T., & Thomas, E. L. (2007). Broad-wavelength-range chemically tunable block-copolymer photonic gels. *Nature Materials*, *6*, 957–960.
7. Woodruff, M. A., & Hutmacher, D. W. (2010). The return of a forgotten polymer — Polycaprolactone in the 21st century. *Progress in Polymer Science*, *35*, 1217–1256.
8. Westbrook, K. K., & Qi, H. J. (2008). Actuator designs using environmentally responsive hydrogels. *Journal of Intelligent Material Systems and Structures*, *19*(5), 597–607.
9. Yakacki, B. C. M., Shandas, R., Safranski, D., Ortega, A. M., Sassaman, K., & Gall, K. (2008). Strong , Tailored , Biocompatible Shape-Memory Polymer Networks. *Advanced Functional Materials*, *18*, 2428–2435.

10. Anand, L., Ames, N. M., Srivastava, V., & Chester, S. A. (2009). A thermo-mechanically coupled theory for large deformations of amorphous polymers. Part I: Formulation. *International Journal of Plasticity*, *25*(8).
11. Hu, J., Meng, H., Li, G., & Ibekwe, S. I. (2012). A review of stimuli-responsive polymers for smart textile applications. *IOP Science*, *21*(053001).
12. Lebel, L. L., Aissa, B., Ali, M., Khakani, E., & Therriault, D. (2010). Ultraviolet-Assisted Direct-Write Fabrication of Carbon Nanotube / Polymer Nanocomposite Microcoils. *Advanced Materials*, *22*, 592–596.
13. Roy, D., Cambre, J. N., & Sumerlin, B. S. (2010). Future perspectives and recent advances in stimuli-responsive materials. *Progress in Polymer Science*, *35*, 278–301.
14. Stuart, M. A. C., Huck, W. T. S., Genzer, J., Müller, M., Ober, C., Stamm, M., … Minko, S. (2010). Emerging applications of stimuli-responsive polymer materials. *Nature Materials*, *9*(2).
15. Zarek, M., Layani, M., Eliazar, S., Mansour, N., Cooperstein, I., Shukrun, E., … Magdassi, S. (2016). 4D printing shape memory polymers for dynamic jewellery and fashionwear. *Virtual and Physical Prototyping*, *11*(4), 263–270.
16. Ahmed, S., Lauff, C., Ounaies, Z., & Simpson, T. W. (2013). Multi-Field Responsive Origami Structures : Preliminary Modeling and Experiments. *ASME 2013 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference IDETC/CIE 2013*, (August).
17. Holmes, D. P., Roché, M., Sinha, T., & Stone, H. A. (2011). Bending and twisting of soft materials by non-homogenous swelling. *Soft Matter*, *7*(11), 5188–5193.
18. Best, J. P., Neubauer, M. P., Javed, S., Dam, H. H., Fery, A., & Caruso, F. (2013). *Mechanics of pH-Responsive Hydrogel Capsules*.
19. Stoychev, G., Turcaud, S., Dunlop, J. W. C., & Ionov, L. (2013). Hierarchical Multi-Step Folding of Polymer Bilayers. *Advanced Functional Materials*, 2295–2300. https://doi.org/10.1002/adfm.201203245
20. Thérien-aubin, H., Wu, Z. L., Nie, Z., & Kumacheva, E. (2013). Multiple Shape Transformations of Composite Hydrogel Sheets Multiple. *Journal of the American Chemical Society*.
21. Knaian, A. N. (2013). Programmable matter. *Physics Today*, *66*(6), 64–65.
22. Ge, Q., Qi, H. J., Dunn, M. L., Ge, Q., Qi, H. J., & Dunn, M. L. (2013). Active materials by four-dimension printing. *Applied Physics Letters*, *103*(131901), 102–108.
23. Tibbits, S. (2013). *4D Printing: Multi Material Shape Change*. 116–121.
24. Lee, Y., Lee, H., Hwang, T., Lee, J., & Cho, M. (2015). Sequential Folding using Light- activated Polystyrene Sheet. *Nature Publishing Group*, 1–9.
25. Khare, V., Sonkaria, S., Lee, G., Ahn, S., & Chu, W. (2017). *From 3D to 4D Printing – Design , Material and Fabrication for Multi-Functional Multi-Materials*. (July).
26. Khoo, Z. X., Ee.Mei.Teoh, J., Liu, Y., Chua, C. K., Yang, S., An, J., … Yeong, W. Y. (2015). 3D printing of smart materials : A review on recent progresses in 4D printing. *Virtual and Physical Prototyping*, *10*(3), 103–122.
27. Raviv, D., Zhao, W., McKnelly, C., Papadopoulou, A., Kadambi, A., Shi, B., … Olguin, C. (2014). Active printed materials for complex self evolving deformations. *Sci. Rep.*, *4*(7422), 1–9.
28. Ge, Q., Dunn, C. K., Qi, H. J., & Dunn, M. L. (2014). Active origami by 4D printing. *IOP Science*, *094007*.
29. Lauff, C., Simpson, T. W., Frecker, M., Ounaies, Z., Ahmed, S., Lockette, P. Von, … Lien, J. (2014). DIFFERENTIATING BENDING FROM FOLDING IN ORIGAMI ENGINEERING USING ACTIVE MATERIALS Carlye. *Proceedings of the ASME 2014 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference IDETC/CIE 2014*, 1–12. Buffalo, New York, USA.
30. Naficy, S., Gately, R., Iii, R. G., Xin, H., & Spinks, G. M. (2016). 4D Printing of Reversible Shape Morphing Hydrogel Structures. *Macromolecular Materials and Engineering*, 1–9.
31. Peraza-hernandez, E. A., Hartl, D. J., & Jr, R. J. M. (2014). Origami-inspired active structures : a synthesis and review. *IOP Science*, *094001*.
32. Hager, M. D., Bode, S., Weber, C., & Schubert, U. S. (2015). Shape memory polymers : Past , present and future developments. *Progress in Polymer Science*, *49–50*, 3–33.
33. Khademhosseini, A., & Langer, R. (2016). A decade of progress in tissue engineering. *10 Years of Nature Protocols*, *11*(10), 6–9.
34. Sydney Gladman, A., Matsumoto, E. A., Nuzzo, R. G., Mahadevan, L., & Lewis, J. A. (2016). Biomimetic 4D printing. *Nature Materials*, *15*(4), 413–418.
35. Sossou, G., Demoly, F., Belkebir, H., Qi, H. J., Gomes, S., & Montavon, G. (2019a). Design for 4D printing : A voxel-based modeling and simulation of smart materials. *Materials & Design*, *175*, 107798.
36. Sossou, G., Demoly, F., Belkebir, H., Qi, H. J., Gomes, S., & Montavon, G. (2019b). Design for 4D printing: Modeling and computation of smart materials distributions. *Materials and Design*, *181*.
37. Borst, W. (1997). Construction of Engineering Ontologies.
38. Gruber, T. R. (1993). A Translation Approach to Portable Ontology Specifications. *Knowledge Aquisition*, *5*(2), 199–220.
39. Studer, R., Benjamins, V. R., & Fensel, D. (1998). Knowledge Engineering: Principles and methods. *Data and Knowledge Engineering*, *25*(1–2), 161–197.
40. Dartigues, C., Ghodous, P., Gruninger, M., Pallez, D., & Sriram, R. (2007). CAD / CAPP Integration Using Feature Ontology. *Concurrent Engineering: Research and Applications*, *15*(2).
41. Grüninger, M., & Delaval, A. (2019). A First-Order Cutting Process Ontology for Sheet Metal Parts. *Frontiers in Artificial Intelligence and Applications, Formal Ontologies Meet Industry 198 IOS Press*, *8*, 22–33. Amsterdam.

42. Bateman, J., Garcia, A., & Garcia, L. E. R. (2011). An Ontology-Based Feature Racognition and Design Rule Checker for Engineering. *Proceedings of the Workshop "Ontologies Come of Age in the Semantic Web,"* 48–59. Bonn, Germany.
43. Kim, S., Rosen, D. W., Witherell, P., & Ko, H. (2019). A Design for Additive Manufacturing Ontology to Support Manufacturability Analysis. *Journal of Computing and Information Science in Engineering*, *19*(4).
44. Liang, J. S. (2018). An ontology-oriented knowledge methodology for process planning in additive layer manufacturing. *Robotics and Computer-Integrated Manufacturing*, *53*(March), 28–44.
45. Zhang, Y., Luo, X., Zhao, Y., Zhang, H.C., 2015b. An ontology-based knowledge frame-work for engineering material selection. Adv. Eng. Inform. 29 (4), 985–1000.
46. Hagedorn, T. J., Smith, B., Krishnamurty, S., Grosse, I., Hagedorn, T. J., Smith, B., … Grosse, I. (2019). *Interoperability of disparate engineering domain ontologies using basic formal ontology ABSTRACT*. *4828*.
47. Dimassi, S., Demoly, F., Cruz, C., Qi, H.J., Kin, K.Y., Andre, J.C., Gomes, S., An ontology based-framework to formalize and represent 4D printing knowledge in design (2021). Computers in Industry.
48. Baader, F., Horrocks, I., Lutz, C., & Sattler, U. (2017). *An Introduction to description Logic*.
49. Andreasen, M. M. (2011). 45 Years with design methodology. *Journal of Engineering Design*, *22*(5), 293–332.
50. Liu, J., & Hu, X. (2013). A reuse oriented representation model for capturing and formalizing the evolving design rationale. *Artificial Intelligence for Engineering Design, Analysis and Manufacturing: AIEDAM*, *27*(4), 401–413.
51. Bonvoisin, J., Halstenberg, F., Buchert, T., & Stark, R. (2016). A systematic literature review on modular product design. *Journal of Engineering Design*, *27*(7), 488–514.
52. Demoly, F., Monticolo, D., Eynard, B., Rivest, L., & Gomes, S. (2010). Multiple viewpoint modelling framework enabling integrated product-process design. *International Journal on Interactive Design and Manufacturing*, *4*(4), 269–280.
53. Demoly, F., Yan, X. T., Eynard, B., Rivest, L., & Gomes, S. (2011). An assembly oriented design framework for product structure engineering and assembly sequence planning. *Robotics and Computer-Integrated Manufacturing*, *27*(1), 33–46.
54. Kuo, T. C., Huang, S. H., & Zhang, H. C. (2001). Concurrent engineering and DFMA/DFX in the development of automotive components. *Procedia CIRP*, *41*(May), 241–260.
55. Segonds, F., Rivette, M., Turpault, S., & Peyre, P. (2019). Design for additive manufacturing (DfAM) methodologies: a proposal to foster the design of microwave waveguide components. *Virtual and Physical Prototyping*, *14*(2), 175–187.
56. Wiberg, A., Persson, J., & Ölvander, J. (2019). Design for additive manufacturing – a review of available design methods and software. *Rapid Prototyping Journal*, *25*(6), 1080–1094.